



# Image Forgery Detection Using Error Level Analysis and Multi-Scale CNN

## CASIA 2.0 Dataset

### Abstract

Digital image forgery has become a critical threat to information integrity across journalism, legal evidence, and social media. This paper proposes a Multi-Scale Convolutional Neural Network (CNN) enhanced with Squeeze-and-Excitation (SE) attention for binary image forgery detection. The proposed model leverages Error Level Analysis (ELA) as a preprocessing technique to amplify compression inconsistencies introduced by tampering operations such as splicing and copy-move. The model adopts an Inception-style multi-scale feature extraction strategy that simultaneously captures fine-grained pixel-level ELA artifacts, local texture patterns, and coarse tampered regions using parallel 1x1, 3x3, and 5x5 convolutional branches. The architecture is trained and evaluated on the CASIA 2.0 Image Tampering Detection Dataset. Experimental results demonstrate that the ELA-based pipeline achieves a test accuracy of 92.08%, outperforming the baseline model trained on raw original images which achieved 88.06%. The proposed approach offers a computationally efficient and effective framework for real-world digital forensics applications.

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**Keywords:** *Image Forgery Detection, Error Level Analysis, Multi-Scale CNN, Squeeze-and-Excitation, CASIA 2.0, Deep Learning, Digital Image Forensics, Inception Block*

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## 1. Introduction

Digital photography and image editing software have become ubiquitous in daily life, enabling the creation of highly realistic forged images with minimal technical expertise. Image forgeries, including splicing, copy-move, and retouching, pose substantial threats to domains where visual authenticity is paramount, such as digital journalism, legal proceedings, medical imaging, and forensic investigations [1]. The rapid proliferation of social media platforms further amplifies the risk of manipulated content spreading unchecked, potentially causing misinformation and public harm [2].

Image forgery detection, a core component of digital image forensics, aims to identify manipulated regions within a given image. Traditional approaches based on handcrafted features, such as Discrete Cosine Transform (DCT) and Local Binary Patterns (LBP), have been widely studied but often struggle to generalize across diverse tampering types and image qualities [3]. The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly advanced the field, enabling automatic extraction of discriminative features directly from image data [4].

Error Level Analysis (ELA) is a widely adopted forensic technique that exploits compression artifacts introduced during image manipulation. When a digital image is saved in JPEG format, each region undergoes uniform compression. If a region has been modified and re-saved, it



exhibits a different error level compared to unaltered regions. By re-compressing the image and computing the pixel-wise difference, ELA effectively highlights tampered areas as bright regions, making them visually distinguishable from authentic content [5].

In this work, we propose a Multi-Scale CNN framework that integrates ELA preprocessing with an Inception-style architecture enhanced by Squeeze-and-Excitation (SE) channel attention. The model captures ELA artifacts simultaneously at three spatial scales (1x1, 3x3, and 5x5 convolution kernels), enabling it to detect both fine-grained pixel-level manipulation traces and broader regional inconsistencies. We conduct comprehensive experiments on the CASIA 2.0 dataset, comparing ELA-based and raw image-based training pipelines. Our contributions are as follows:

1. We integrate ELA as a preprocessing step with a custom multi-scale deep learning architecture, demonstrating its superiority over raw image training for forgery detection.
2. We propose an Inception-style multi-scale feature extraction block that captures ELA artifacts at multiple granularities, combined with SE attention for adaptive channel recalibration.
3. We provide a thorough comparative analysis between ELA-based and original-image pipelines on the CASIA 2.0 benchmark dataset, achieving 92.08% accuracy with the ELA pipeline.

Despite significant progress, key challenges remain: ELA effectiveness is limited on re-compressed or lossless images; most models lack cross-dataset generalization beyond CASIA; binary detection methods do not provide spatial forgery localization; and the rise of AI-generated imagery introduces forgery types that bypass traditional JPEG-based detection [3][4].

## 2. Related Work

The detection of image forgeries has been studied extensively using both traditional signal processing techniques and modern deep learning approaches. This section reviews the most relevant prior works in image forgery detection, ELA-based methods, and attention-enhanced deep learning architectures.

In [1] A CNN-based framework was proposed a CNN-based framework that combines ELA with a shallow two-layer convolutional network for binary classification of spliced and copy-move forgeries. The method demonstrated that ELA preprocessing significantly boosts CNN classification performance by providing forensic-sensitive feature maps, achieving competitive results on multiple public datasets. Their work formed a foundational baseline for ELA-guided deep learning forgery detection. The method achieved a test accuracy of 94.14% on the CASIA 2.0 dataset [1].

In [3] ELA was integrated with a custom CNN for image tampering detection, using ELA representations preprocessed from the CASIA v2.0 dataset to train a classifier that distinguishes authentic from manipulated regions. The framework outperformed established deep learning baselines including VGG16, VGG19, and ResNet101, achieving 96.21% detection accuracy on CASIA v2.0 [3].



In [4] ELA-Conv was proposed, integrating Error Level Analysis with a convolutional neural network for binary image forgery detection. The method attains a detection accuracy of 94% and an F-score of 94%, outperforming pre-trained baseline models including VGG16 and VGG19 on the CASIA dataset [4].

In [6] ELA was applied ELA with prominent transfer learning architectures including VGG-16, VGG-19, ResNet-50, and Xception for image forgery detection on the CASIA 1.0 dataset. Their work demonstrated that combining ELA preprocessing with pretrained CNN backbones yields superior accuracy compared to training directly on original images, with ELA showing superiority over patch-level techniques in capturing subtle manipulation artifacts [6]. The proposed ELA-CNN framework achieved a detection accuracy of 96.21% on the CASIA v2.0 dataset [6].

In [7] DMOBILE-ELA was proposed DMOBILE-ELA, a cascaded model that combines Atrous MobileNet with ELA for digital image forgery detection. The cascaded design enabled the model to extract multi-receptive-field features from ELA-preprocessed images, improving detection of both localized and diffuse manipulation artifacts. Experiments showed strong performance on the CASIA benchmark [7]. The DMOBILE-ELA model achieved a validation accuracy of 98.48% on the CASIAv2.0 dataset [7].

In [8] A method to detect splicing forgery was proposed a method to detect splicing image forgery using a custom CNN architecture evaluated on the CASIA v2.0 dataset. The approach extracted hierarchical forensic features through multiple convolutional depths and demonstrated strong detection performance against various splicing manipulations including boundary inconsistencies and compression artifact mismatches. The proposed CNN achieved 99.3% accuracy on the CASIA v2.0 dataset for splicing forgery detection [8].

In [9] A combined classification and segmentation system was proposed using ELA preprocessing with a VGG16-UNet model for image forgery detection. Evaluated on the CASIA v2.0 dataset, the classification path achieved 91.7% accuracy and 93.52% precision, while the segmentation path yielded an IoU of 80.6% and an F1-score of 77.7% [9].

In [10] A transfer learning method was introduced a method applying transfer learning with MobileNetV2, VGG16, and ResNet50 for binary forgery classification tested directly on the CASIA 2.0 dataset. Among the evaluated models, MobileNetV2 achieved the highest test accuracy of approximately 95%. The study highlighted key trade-offs between detection accuracy, training speed, and computational cost in transfer learning-based detection pipelines.

In [12] A deep learning framework was proposed a deep learning framework for efficient and robust image forgery detection combining transfer learning with particle swarm optimization feature selection and Gated Recurrent Units classification. The architecture was validated on the modified CASIA dataset and demonstrated strong robustness against diverse manipulation types, highlighting the value of optimized hybrid deep learning pipelines [12]. The proposed PSO+GRU framework achieved 96.25% accuracy on the modified CASIA dataset [12].

In [13] A ResNet50-based CNN augmented with Global Average Pooling and Dropout regularization was trained on ELA-transformed images from the CASIA 2.0 dataset. The model uses Adam optimization with binary cross-entropy loss and early stopping, achieving approximately 93% accuracy on CASIA 2.0, demonstrating the effectiveness of combining ELA preprocessing with residual network architectures [13].



In [16] SFTA-Net was proposed SFTA-Net, a self-supervised lightweight CNN for detecting copy-move and splicing forgeries using triplet loss, auxiliary loss, and spatial attention without labeled masks. The model introduces a commutative loss combining classification, auxiliary, and triplet losses to maximize the distance between tampered and original image features, achieving competitive accuracy on the CASIA 2.0 benchmark [16]. SFTA-Net achieved 97% accuracy on the CASIA 2.0 test set [16].

In [17] A Vision Transformer (ViT) was introduced a Vision Transformer (ViT) combined with SVM for dual-category image classification, targeting both copy-move and splicing forgeries. The ViT architecture captures long-range dependencies across the entire image using self-attention, enabling detection of globally distributed manipulation artifacts that are difficult for local receptive field models to capture [17]. The ViT+SVM framework achieved 97.85% accuracy on the CASIA v1.0 dataset [17].

In [18] A hybrid dual-stream CNN approach combined Error Level Analysis with Spatial Rich Model (SRM) noise residuals for image forgery detection on the CASIA 2.0 dataset. The dual-stream design exploits both compression-level artifacts from ELA and high-frequency noise patterns from SRM, achieving 98.55% accuracy on the CASIA 2.0 benchmark [18].

In [19] Multi-resolution transfer learning was applied multi-resolution transfer learning using EfficientNetV2B0 enhanced with Squeeze-and-Excitation (SE) attention for tampered image classification, achieving AUC scores up to 1.0000 on four benchmark datasets. The SE block recalibrates channel-wise feature responses by applying global average pooling followed by a bottleneck fully connected network, producing attention weights that emphasize forensically informative channels. This mechanism demonstrated the most effective SE-guided forgery detection result to date [19]. The SE-enhanced EfficientNetV2B0 model achieved AUC scores of up to 1.00 across four benchmark datasets [19].

In [20] A hybrid pipeline integrating Error Level Analysis, Scale-Invariant Feature Transform (SIFT), and a CNN was proposed for image forgery detection on the CASIA v2.0 dataset. While standalone ELA achieved 78.5% accuracy and the CNN alone reached 95.8%, the integrated hybrid pipeline delivered 97.1% accuracy with improved precision, demonstrating that combining complementary forensic techniques yields superior performance [20].

*Table 1. Summary of Related Research Studies on Image Forgery Detection*

**Table 1. Summary of Related Research Studies on Image Forgery Detection**

| Ref. | Techniques                                       | 3.2 Dataset                       | Accuracy     | Key Contribution                                                                                   |
|------|--------------------------------------------------|-----------------------------------|--------------|----------------------------------------------------------------------------------------------------|
| [1]  | ELA + Shallow CNN (2-layer)                      | CASIA 2.0                         | 94.14%       | ELA + 2-layer CNN; 94.14% test accuracy on CASIA 2.0; first ELA-guided DL forensic baseline.       |
| [3]  | ELA + Custom CNN                                 | CASIA v2.0                        | 96.21%       | ELA + CNN; outperforms VGG16/VGG19/ResNet101; 96.21% on CASIA v2.0.                                |
| [4]  | ELA-Conv (ELA + CNN)                             | CASIA                             | 94%          | ELA-Conv: ELA + CNN; 94% accuracy and F-score; outperforms VGG16/VGG19.                            |
| [6]  | ELA + VGG-16/19, ResNet-50, Xception             | CASIA v2.0                        | 96.21%       | ELA + pretrained CNN backbones; ELA outperforms patch-level techniques on CASIA v2.                |
| [7]  | DMOBILE-ELA (Atrous MobileNet + ELA)             | CASIA v2.0                        | 98.48% (val) | Atrous MobileNet + ELA; multi-receptive-field features; 98.48% val on CASIA v2.                    |
| [8]  | Lightweight Custom CNN (Splicing)                | CASIA v2.0                        | 99.3%        | Lightweight 4-conv-layer CNN for splicing; 99.3% on CASIA v2.0 with minimal parameters.            |
| [9]  | ELA + VGG16-UNet                                 | CASIA v2.0                        | 91.7%        | ELA + VGG16-UNet; classification 91.7%, segmentation IoU 80.6% on CASIA v2.                        |
| [10] | Transfer Learning: MobileNetV2, VGG16, ResNet50  | CASIA 2.0                         | ~95%         | Transfer learning; MobileNetV2 achieves ~95% on CASIA 2.0; accuracy vs. speed trade-off analyzed.  |
| [12] | Transfer Learning + PSO + GRU                    | Modified CASIA                    | 96.25%       | PSO feature selection + GRU classifier; 96.25% on modified CASIA; robust to noisy inputs.          |
| [13] | ELA + ResNet50 + GAP + Dropout                   | CASIA 2.0                         | ~93%         | ELA + ResNet50 + GAP + Dropout; Adam optimizer; ~93% accuracy on CASIA 2.0.                        |
| [16] | SFTA-Net (Self-Supervised + Triplet + Attention) | CASIA 2.0                         | 97%          | Self-supervised CNN; commutative loss (classification + triplet); 97% on CASIA 2.0.                |
| [17] | Vision Transformer (ViT) + SVM                   | CASIA v1.0                        | 97.85%       | ViT + SVM; long-range self-attention; 97.85% on CASIA v1.0; robust against adversarial attacks.    |
| [18] | Dual-stream CNN + ELA + SRM Noise                | CASIA 2.0                         | 98.55%       | Dual-stream CNN combining ELA compression artifacts with SRM noise residuals; 98.55% on CASIA 2.0. |
| [19] | EfficientNetV2B0 + SE Attention + Focal Loss     | CASIA v1, Columbia, MICC, Defacto | AUC 1.00     | EfficientNetV2 + SE + Focal Loss; AUC 1.00 on 4 benchmarks; outperforms 42 SOTA models.            |
| [20] | ELA + SIFT + CNN Hybrid Pipeline                 | CASIA v2.0                        | 97.1%        | ELA + SIFT + CNN hybrid; 97.1% vs. 95.8% CNN-only and 78.5% ELA-only on CASIA v2.                  |





### 3. Proposed Methodology

This section presents the proposed methodology adopted for detecting image forgeries using deep learning techniques. The approach is designed to highlight compression inconsistencies and forensic traces introduced by tampering operations through Error Level Analysis (ELA) before feeding the processed images into a convolutional neural network.

The methodology is structured into three main stages. First, the common types of image forgery and the role of ELA in exposing manipulation artifacts are discussed. Next, the dataset used in this study is described along with its characteristics and splitting strategy. Finally, the preprocessing pipeline, including ELA conversion, resizing, and data augmentation techniques, is detailed to ensure consistent and effective model training.

#### 3.1 Image Forgery Types and ELA

Digital image forgery refers to the intentional manipulation of an image to deceive the observer or misrepresent the depicted content. Forgeries may be broadly categorized into three main types [1][3]:

**Image Splicing:** A region from one image is extracted and pasted into a different target image. This introduces boundary inconsistencies, compression artifacts, and noise pattern mismatches that ELA can detect.

**Copy-Move Forgery:** A region within the same image is copied and pasted to a different location, typically to conceal or duplicate an object. This creates local statistical redundancies detectable through key points and block-based analysis.

**Retouching and Inpainting:** Subtle modifications applied to alter visual content, often through smoothing, removal of objects, or enhancement of specific regions, introducing imperceptible but forensically detectable pixel-level traces.

**Error Level Analysis (ELA):** ELA amplifies JPEG compression inconsistencies by re-saving the image at a fixed quality level (e.g., 90%) and computing the absolute difference between the re-saved image and the original. Authentic regions that have been uniformly saved exhibit uniform, low error level, while tampered regions that have undergone additional saves display elevated error levels, rendered as bright areas in the ELA map. This makes ELA a powerful preprocessing tool for exposing forgery boundaries and manipulated patches [5][6].

#### 3.2 Dataset

The CASIA 2.0 Image Tampering Detection Dataset is a widely used benchmark in digital image forensics [21]. It contains a total of 12,614 images divided into authentic (real) and tampered (fake) categories, encompassing various forgery types including splicing and copy-move. The dataset was sourced from Kaggle (<https://www.kaggle.com/datasets/divg07/casia-20-image-tampering-detection-dataset>) and provides ground truth binary labels for all images. The dataset was split into 70% training, 15% validation, and 15% testing subsets using stratified random splitting with a fixed random seed (42), ensuring no data leakage between splits and preserving class distribution across all subsets.

Table 2 presents a detailed breakdown of the dataset distribution across splits and class categories. Authentic images (Au) were sourced from the CASIA2/Au directory, while tampered images (Tp) were sourced from the CASIA2/Tp directory. The dataset exhibits a moderate class



imbalance, with authentic images comprising approximately 59.4% of the total and tampered images comprising 40.6%, which was addressed during training through class-weighted loss computation.

*Table 2. CASIA 2.0 Dataset Distribution Across Splits and Class Categories.*

| Split      | Authentic (Real) | Tampered (Fake) | Total Images | Proportion (%) |
|------------|------------------|-----------------|--------------|----------------|
| Training   | 5,244            | 3,586           | 8,830        | 70%            |
| Validation | 1,124            | 769             | 1,893        | 15%            |
| Testing    | 1,123            | 768             | 1,891        | 15%            |
| Total      | 7,491 (59.4%)    | 5,123 (40.6%)   | 12,614       | 100%           |

### 3.3 Data Preprocessing and ELA Conversion

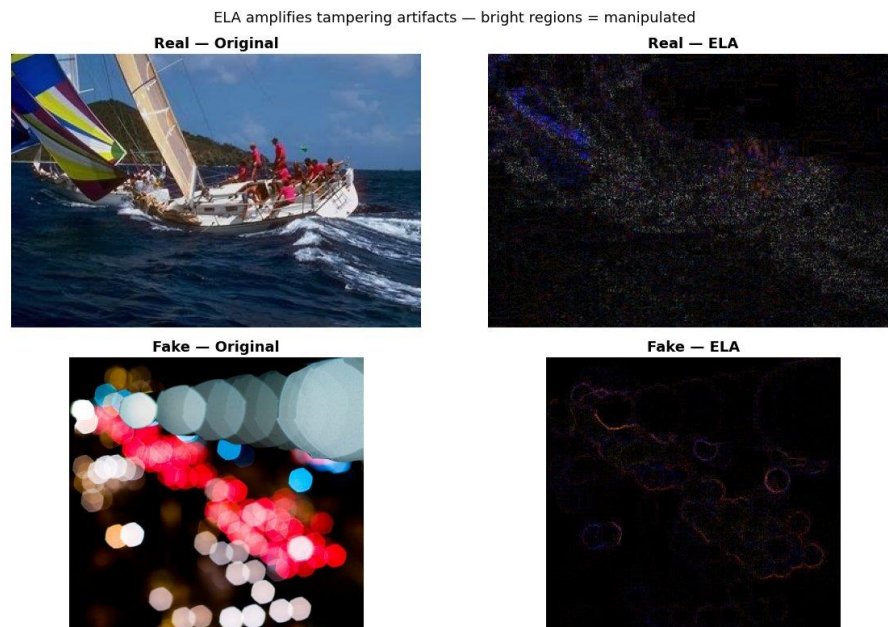
Each image was converted to ELA format as a preprocessing step. The ELA conversion process involves three stages: (1) re-saving the image as JPEG at 90% quality, (2) computing the pixel-wise absolute difference between the original and re-saved images, and (3) amplifying the difference by normalizing the maximum error level to enhance visual contrast. All images were then resized to 128x128 pixels to maintain consistent input dimensions across the CNN. Data augmentation was applied during training including horizontal and vertical flipping, rotation up to 20 degrees, width/height shifts of 10%, zoom range of 15%, shear range of 10%, and brightness variation between 0.75 and 1.25. Class weights were applied to address dataset class imbalance between real and fake categories.

$$ELA(x, y) = \alpha \cdot |I(x, y) - J_q(I)(x, y)| \quad (1)$$

#### Error Level Analysis (ELA)

This equation (1) defines the Error Level Analysis transformation applied to each pixel. The original image  $I(x, y)$  is re-compressed at a fixed JPEG quality  $q$ , and the absolute difference between the original and compressed image is computed. The scaling factor  $\alpha$  enhances contrast, making compression inconsistencies caused by tampering visually and computationally distinguishable for CNN.

Figure 1 and Figure 2 illustrate examples of real and forged images alongside their corresponding ELA representations. Authentic images exhibit uniformly low error levels (dark ELA maps), while forged images display elevated, localized bright regions corresponding to tampered areas.



*Figure 1. ELA amplifies tampering artifacts — bright regions indicate manipulated areas (Set 1).*



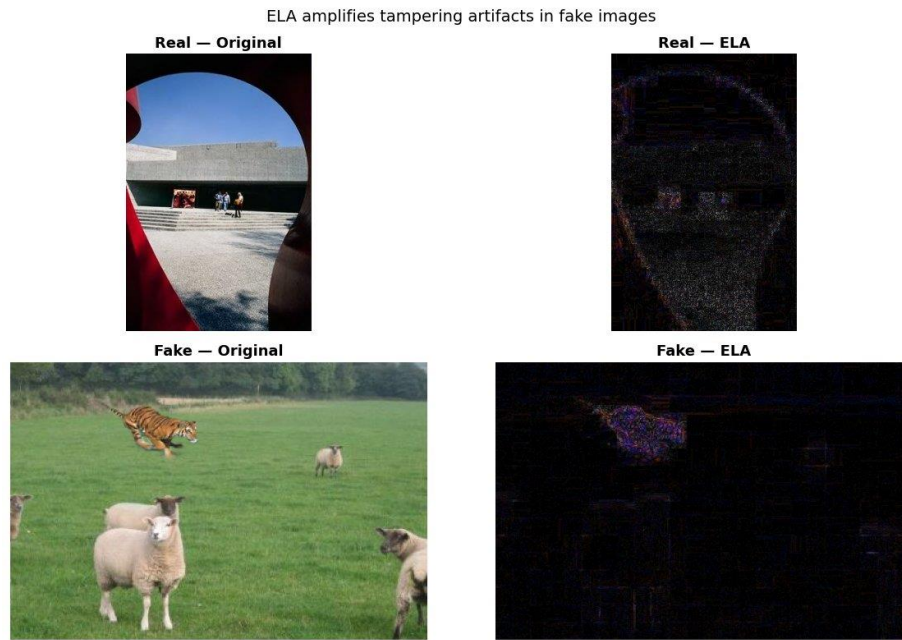


Figure 2. ELA amplifies tampering artifacts in fake images — manipulated objects are clearly highlighted (Set 2).

### 3.4 Model Architecture: Multi-Scale CNN with SE Attention

The proposed model adopts a custom Multi-Scale CNN architecture inspired by the Inception design philosophy. The core building block is a multi-scale Inception-style module that processes the input feature maps through three parallel convolutional branches with kernel sizes of 1x1, 3x3, and 5x5. These branches are designed to capture ELA artifacts at different spatial granularities: the 1x1 branch captures pixel-level compression differences, the 3x3 branch captures local texture patterns, and the 5x5 branch captures coarser regional inconsistencies. The outputs of the three branches are concatenated along the channel dimension, followed by Batch Normalization and ReLU activation.



$$F_k = \sigma(W_k * X + b_k), \quad k \in \{1, 3, 5\}$$

(2)

### Multi-Scale Convolution Branches

This equation (2) represents the convolution operation in each parallel branch of the multi-scale module using kernel sizes  $1 \times 1$ ,  $3 \times 3$ , and  $5 \times 5$ . Each branch extracts features at a different spatial resolution, allowing the network to capture pixel-level, local, and regional forgery artifacts.

$$F_{concat} = \text{Concat}(F_1, F_3, F_5)$$

(3)

### Concatenation of Multi-Scale Features

The outputs of the three convolution branches (3) are concatenated along the channel dimension to form a unified multi-scale feature map. This operation preserves information from all spatial granularities for subsequent processing.

After each multi-scale block, a Squeeze-and-Excitation (SE) block is applied to recalibrate the concatenated channel features. The SE block employs Global Average Pooling to produce a channel descriptor, followed by a two-layer fully connected bottleneck with ReLU and Sigmoid activations to generate channel-wise attention weights. These weights multiplicatively scale the original feature maps, suppressing uninformative channels and emphasizing forensically relevant features.

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{concat}^{(c)}(i, j)$$

(4)

### Squeeze Operation (Global Average Pooling)

This equation (4) computes the channel-wise descriptor by averaging spatial values across the feature map. It compresses spatial information into a single representative value per channel, preparing the features for attention weighting.

$$s = \sigma (W_2 \cdot \delta (W_1 \cdot z))$$

(5)

#### Excitation Operation (SE Bottleneck)

The excitation step passes the channel descriptor through a bottleneck of two fully connected layers with ReLU and Sigmoid activations. This produces attention weights that determine the importance of each feature channel.

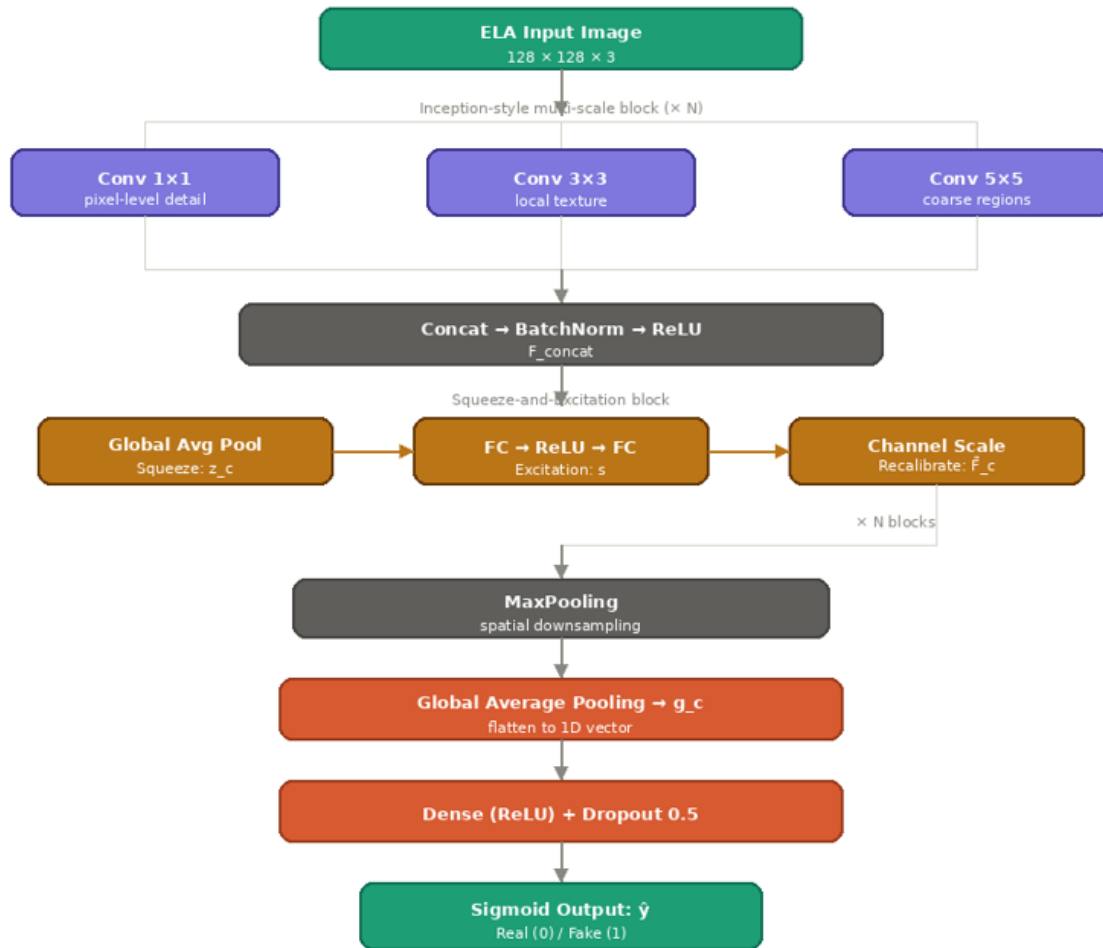
$$\tilde{F}^{(c)} = s_c \cdot F_{concat}^{(c)}$$

(6)

#### Channel Recalibration

Each channel of the concatenated feature map is multiplied by its corresponding attention weight. This suppresses less informative channels and emphasizes channels containing strong forensic signals.

The overall network stacks multiple such blocks with interleaved MaxPooling layers for spatial downsampling, followed by Global Average Pooling, a Dense layer with ReLU activation, a Dropout layer (rate = 0.5) for regularization, and a final sigmoid output neuron for binary classification. The model is compiled using the AdamW optimizer with a learning rate of  $8 \times 10^{-4}$  and weight decay of  $1 \times 10^{-4}$ , binary cross-entropy loss with label smoothing of 0.05, and a warm-up cosine learning rate schedule over 60 epochs. Early stopping is applied with a patience of 10 epochs while monitoring validation accuracy. **The complete architectural design of the proposed Multi-Scale CNN with Squeeze-and-Excitation attention is illustrated in Figure 3.**



**Figure 3. Architecture of the proposed Multi-Scale CNN with Squeeze-and-Excitation Attention for image forgery detection.**

## 4. Experimental Results

We conducted a comparative evaluation of the proposed Multi-Scale CNN across two training pipelines: one trained on ELA-preprocessed images and one trained on the original raw images. Both pipelines share the identical model architecture, optimizer configuration, and dataset split to ensure a fair comparison. Model performance was evaluated using test accuracy, recall, precision, F1-score, and confusion matrix analysis.

### 4.1 Performance Comparison: ELA vs. Original

Table 3 summarizes the quantitative performance of both pipelines on the CASIA 2.0 test set. The ELA-based model consistently outperforms the original image model across all metrics. The ELA pipeline achieved a test accuracy of 92.08%, a 4.02 percentage point improvement over the 88.06% obtained by the raw image baseline.

#### Evaluation Metrics

The model performance is evaluated using standard classification metrics derived from the confusion matrix, including Accuracy, Precision, Recall, and F1-Score. Where TP (True Positive) and TN (True Negative) denote correctly classified forged and authentic images, respectively, while FP (False Positive) and FN (False Negative) represent misclassifications.

#### Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

#### Precision

$$Precision = \frac{TP}{TP + FP}$$

#### Recall

$$Recall = \frac{TP}{TP + FN}$$

#### F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$



Table 3 summarizes the quantitative performance of both pipelines on the CASIA 2.0 test set.

**Table 3. Performance Comparison on CASIA 2.0 Test Set**

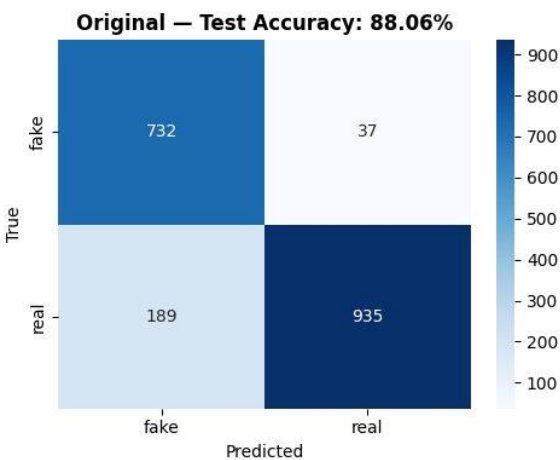
| Model                                  | Accuracy      | Precision    | Recall       | F1-Score     |
|----------------------------------------|---------------|--------------|--------------|--------------|
| <b>Multi-Scale CNN + SE (Proposed)</b> | <b>92.08%</b> | <b>93.7%</b> | <b>93.7%</b> | <b>92.1%</b> |
| Multi-Scale CNN + SE (Baseline)        | 88.06%        | 95.2%        | 83.2%        | 88.9%        |

#### Confusion Matrix Analysis

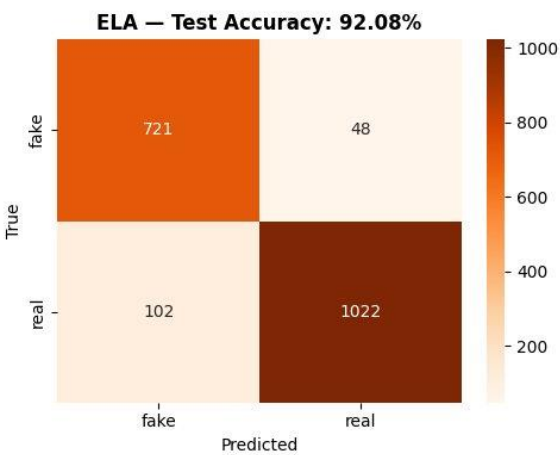
**Figure 4** presents the confusion matrix of the baseline model trained on original images (88.06% accuracy). The model correctly classified 732 fake and 935 real images, while misclassifying 189 real images as fake and 37 fake images as real. The relatively high false negative rate for real images (189 misclassifications) indicates the model struggles to distinguish authentic images when trained on raw pixel features alone.

**Figure 5** presents the confusion matrix of the ELA-based model (92.08% accuracy). The ELA model correctly classified 721 fake and 1022 real images, with only 102 real images misclassified as fake and 48 fake images misclassified as real. The significant reduction in false negatives (from 189 to 102) demonstrates that ELA preprocessing substantially improves the model's ability to correctly identify authentic images, a critical requirement in forensic applications.





*Figure 4. Confusion Matrix — Original Images (88.06%).*

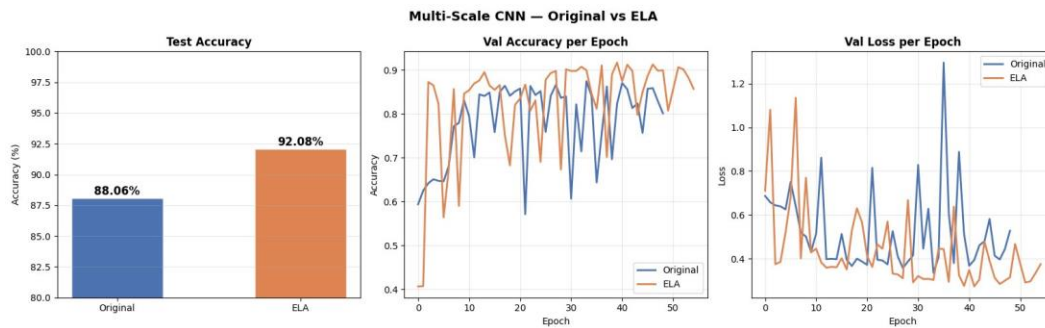


*Figure 5. Confusion Matrix — ELA Images (92.08%).*



#### 4.2 Training Dynamics: Accuracy and Loss Curves

Figure 6 illustrates the comparative training curves of both models over 50 epochs. The ELA model (orange) demonstrates notably faster convergence from epoch 1, reaching above 85% validation accuracy within the first 5 epochs due to the informative ELA feature representation. While both models exhibit oscillatory behavior characteristic of AdamW with cosine decay scheduling, the ELA model maintains consistently higher validation accuracy and lower validation loss throughout training, confirming the superiority of ELA as a forensic preprocessing strategy.



**Figure 6. Multi-Scale CNN performance comparison: Test Accuracy (left), Validation Accuracy per Epoch (center), and Validation Loss per Epoch (right) for Original vs. ELA pipelines.**



## 5. Discussion

The experimental findings show that Error Level Analysis provides a strong forensic signal that significantly enhances the CNN's capability to discriminate between forged and authentic images. The observed 4.02% improvement in test accuracy highlights how ELA amplifies subtle manipulation artifacts that are not easily captured when training directly on raw pixel data. By transforming compression inconsistencies into visually distinguishable patterns, ELA supplies the network with features that are inherently aligned with the nature of image tampering. The proposed multi-scale Inception-style architecture is particularly effective in this context because forgery artifacts appear at different spatial resolutions. The  $1 \times 1$  convolutional branch captures fine-grained pixel-level inconsistencies, while the  $3 \times 3$  and  $5 \times 5$  branches aggregate local textures and broader regional anomalies. This multi-granularity feature extraction closely reflects the spatial diversity of manipulation traces, ranging from boundary-level edits to large spliced regions, allowing the model to learn a richer and more comprehensive forensic representation. In addition, the Squeeze-and-Excitation attention mechanism plays a crucial role by dynamically recalibrating channel-wise features. By suppressing less informative channels and emphasizing those that contain strong forensic cues, the attention block reduces background noise in the ELA maps and directs the learning process toward the most discriminative signals. This selective emphasis improves the robustness and reliability of the learned representations. Compared to the baseline model trained on original images, the ELA-based pipeline shows a marked improvement in correctly identifying authentic images, reducing false negatives from 189 to 102. In practical forensic scenarios, minimizing the misclassification of genuine images is essential due to the potential legal and ethical implications. The more balanced confusion matrix achieved by the ELA pipeline therefore underscores its suitability for real-world deployment in digital forensic applications.



## 6. Conclusion & Future Work

In this project, we have worked on the introduction of Multi-Scale CNN architecture enhanced with Squeeze-and-Excitation attention for effective digital image forgery detection using Error Level Analysis as a forensic preprocessing step on the CASIA 2.0 dataset. We have designed and trained deep learning model capable of identifying Copy-Move, Splicing and Editing within a unified model, unlike most existing detectors that focus on a single manipulation type. We have worked on multiple generalized images from various categories such as animal, architecture, art, scene, nature, plant, texture, character, and indoor scene, as this dataset has 7492 Authentic and 5125 Tampered images. With the integrating between ELA and multi-scale feature we have reached out to high accuracy of 92.08% compared to the baseline that was trained on the original images by 88.06%, so this proves that ELA works well at showing the compression differences caused by image tampering as we have gained 4.02% better performance. We are seeking to improve our project and train it on identifying where specifically the tampering occurred in the image by using segmentation-based architecture such as U-Net or Mask R-CNN to produce pixel-level tampering maps, in addition to applying multi-class classification model that could provide more forensic detail by giving what is the type of forgery that happened without depending only on the binary outcome. We would like to keep our eye as well on Real-world tampered images which are used on various social media platforms that can weaken ELA artifacts, so we can handle that by improving Robustness Against Post-Processing model under these conditions.

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